BINARY CLASSIFICATION OF LIVER DISEASE USING MACHINE LEARNING TECHNIQUE

Winter internship project report

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CERTIFICATE

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Signature of Supervisor
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ABSTRACT

Liver disease has become a serious health concern for people all around the world. Factors like unhealthy lifestyles, alcohol use, infections, and poor diet have led to a noticeable rise in liver-related illnesses in recent years. Often, the early signs of liver disease are so mild—or go completely unnoticed—that it's only discovered when it's already advanced. That's why being able to predict and diagnose liver disease early is crucial for better treatment and prevention.

This project set out to create a machine learning system that can predict if someone has liver disease, simply by looking at some basic medical information. The system considers details like age, gender, bilirubin levels, liver enzyme values, and protein levels to make a yes-or-no prediction. For training and testing, it used real-world data from the Indian Liver Patient Dataset.

Before training the model, several data preprocessing steps were carried out, including handling missing values, converting categorical data into numerical form, and applying feature scaling to ensure uniformity among features. Multiple machine learning algorithms were studied and evaluated. Based on performance metrics such as accuracy, recall, and F1-score, Logistic Regression was selected as the final model due to its simplicity, interpretability, and balanced prediction capability.

To enhance usability, a user-friendly web interface was developed using Streamlit. This interface allows users to input patient details and receive instant prediction results. The developed system demonstrates reliable performance and can assist in early identification of liver disease. Although the system is not intended to replace professional medical diagnosis, it can act as a supportive decision-making tool and help raise awareness regarding liver health

1.0 INTRODUCTION

In recent years, the healthcare sector has witnessed rapid technological advancement with the integration of artificial intelligence and machine learning. These technologies have the ability to analyze large volumes of medical data and assist in early disease detection. Machine learning techniques are increasingly being used to support healthcare professionals by providing data-driven insights and improving diagnostic accuracy.

The liver is one of the most important organs of the human body, performing essential functions such as detoxification, metabolism, digestion, and protein synthesis. Liver-related disorders such as hepatitis, fatty liver disease, cirrhosis, and liver failure have become common due to factors including alcohol consumption, unhealthy diet, obesity, viral infections, and lifestyle changes. A major challenge associated with liver disease is that its early symptoms are often mild or unnoticed, leading to delayed diagnosis and increased health risks.

Conventional methods of diagnosing liver disease involve laboratory blood tests, imaging techniques, and expert medical evaluation. Although effective, these methods can be time-consuming and require significant medical resources. In many cases, especially in areas with limited healthcare facilities, early diagnosis becomes difficult. Therefore, there is a need for intelligent systems that can assist in preliminary screening and early prediction of liver disease.

Machine learning offers an efficient approach to this problem by learning patterns from historical patient data. By analyzing medical parameters such as age, bilirubin levels, liver enzyme values, and protein concentrations, machine learning models can classify patients into disease and non-disease categories. Binary classification techniques are particularly suitable for this task as they provide clear and interpretable outcomes.

This project focuses on developing a machine learning–based liver disease prediction system using supervised learning techniques. The system processes patient medical data, applies necessary preprocessing steps, and predicts whether an individual is suffering from liver disease or not. The proposed solution aims to support early diagnosis, increase awareness, and demonstrate the practical application of machine learning in healthcare.

2. LITERATURE SURVEY :

2.1 Overview of Liver Disease Prediction Using Machine Learning

Liver disease prediction aims to identify whether a patient is suffering from liver-related disorders based on clinical and biochemical test parameters. Over the past two decades, healthcare analytics has evolved from manual rule-based diagnosis toward data-driven machine learning approaches. Traditional diagnosis relies heavily on physician expertise and laboratory interpretation, whereas modern systems attempt to learn disease patterns automatically from historical patient data.

Recent research shows that machine learning techniques can significantly improve early detection of liver disease by analyzing attributes such as bilirubin levels, liver enzymes, protein concentration, age, and gender. These systems assist clinicians by providing preliminary screening results and supporting decision-making. A comprehensive review of liver disease prediction techniques can be found in Kodinariya and Gondaliya (2021).

Link:
https://stmcomputers.stmjournals.com/index.php/JoAIRA/article/view/489

2.2 Statistical and Rule-Based Approaches

What they do:
Early liver disease prediction systems relied on statistical thresholds and medical rules defined by domain experts. For example, abnormal enzyme values beyond fixed limits were used to indicate liver dysfunction.

Strengths:

- Simple to implement
- High interpretability
- Useful for basic screening

Limitations:

- Cannot model complex relationships between multiple parameters
- Poor adaptability to diverse patient populations

These limitations motivated researchers to adopt learning-based approach

2.3 Classical Machine Learning Methods

Approaches:
Classical supervised learning algorithms such as Logistic Regression, Support Vector Machines (SVM), Decision Trees, k-Nearest Neighbors, and Random Forest have been widely applied to liver disease datasets. Features typically include blood test results and demographic information.

Strengths:

- Effective with small to medium-sized datasets
- Relatively interpretable

- Computationally efficient

Limitations:

- Performance depends heavily on feature engineering
- Limited ability to capture nonlinear interactions

Several studies using the Indian Liver Patient Dataset (ILPD) demonstrated that Random Forest and Logistic Regression provide stable baseline performance.

Reference:
https://drpress.org/ojs/index.php/HSET/article/view/11926

2.4 Ensemble Learning Approaches

Approaches:
Ensemble models combine multiple classifiers to improve predictive performance. Techniques such as Random Forest, Gradient Boosting, AdaBoost, and voting classifiers have been applied in liver disease prediction.

Strengths:

- Improved accuracy and robustness
- Reduced model variance
- Better generalization

Limitations:

- Increased computational complexity
- Reduced interpretability

Recent research indicates that ensemble methods often outperform single classifiers in clinical prediction tasks.

Link:
https://bmcmedinformdecismak.biomedcentral.com/articles/10.1186/s12911-024-02550-y

2.5 Deep Learning Approaches

Approaches:
Deep learning models such as Artificial Neural Networks (ANN) and Multilayer Perceptrons (MLP) have been explored for liver disease detection. These models automatically learn hidden patterns from numerical medical data.

Strengths:

- Ability to model complex nonlinear relationships
- Higher predictive capability with sufficient data

Limitations:

- Require larger datasets
- Less interpretable than classical ML models
- Higher computational cost

Several studies report improved performance using neural networks, though interpretability remains a challenge in medical applications.

Reference:
https://www.ijraset.com/research-paper/liver-disease-prediction-using-deep-learning

2.6 Dataset-Based Studies and Benchmarking

Most liver disease prediction research utilizes publicly available datasets, particularly the Indian Liver Patient Dataset (ILPD) from the UCI Machine Learning Repository. This dataset contains patient demographic information and biochemical test results.

Researchers have used ILPD extensively for benchmarking algorithms and comparing classification performance across studies.

Dataset link:
https://archive.ics.uci.edu/ml/datasets/Indian+Liver+Patient+Dataset

2.7 Transfer Learning, Model Optimization and Future Trends

Recent studies focus on improving prediction reliability using feature selection, dimensionality reduction, and hyperparameter optimization techniques. Although transfer learning is more common in image-based liver diagnosis, emerging research explores hybrid systems combining clinical data with imaging features.

Future trends emphasize explainable AI (XAI) to ensure trust, transparency, and clinical acceptance of machine learning-based diagnostic systems.

3.1 Problem Definition

Liver disease has emerged as one of the most critical health challenges worldwide due to its increasing occurrence and potentially life-threatening consequences. The liver plays a vital role in essential bodily functions such as metabolism, detoxification, digestion, and protein synthesis. Any impairment in liver function can significantly affect overall human health. Diseases such as hepatitis, fatty liver disease, cirrhosis, and liver failure often develop gradually and may not present clear symptoms during their early stages.

One of the major challenges in liver disease management is early detection. In many cases, patients remain unaware of the condition until the disease has reached an advanced stage. Conventional diagnostic procedures rely on laboratory blood tests, imaging techniques, and expert clinical judgment. Although these methods are effective, they are often time-consuming, expensive, and require specialized medical expertise. Moreover, interpreting multiple biochemical parameters simultaneously can be complex and increases the possibility of human error.

With the rapid growth of digital healthcare systems, large volumes of patient medical data are being generated daily. However, this data is not always effectively utilized for predictive analysis. Traditional diagnostic approaches are limited in their ability to identify hidden patterns and complex relationships among clinical attributes such as bilirubin levels, liver enzyme values, protein concentrations, age, and gender.

Machine learning offers a promising solution to this problem by enabling automated analysis of medical data and learning meaningful patterns from historical records. By applying supervised learning techniques, it is possible to build predictive models that can classify patients into liver disease and non-liver disease categories.

The core problem addressed in this project is the development of a reliable machine learning-based binary classification system capable of predicting liver disease using patient clinical data. The objective is to improve early screening, support medical decision-making, and demonstrate the practical application of artificial intelligence in healthcare diagnostics.

3.2 Scope of the Problem

The scope of this project is centered on designing and implementing a machine learning–based system for predicting liver disease at an early stage using patient clinical data. The primary objective is to develop a binary classification model that can distinguish between individuals who are affected by liver disease and those who are not.

The system makes use of structured medical attributes obtained from routine liver function tests. These attributes include demographic details such as age and gender, along with biochemical parameters such as total bilirubin, direct bilirubin, alkaline phosphatase, alanine aminotransferase (ALT), aspartate aminotransferase (AST), total proteins, albumin, and albumin–globulin ratio. These parameters are commonly available in hospitals and diagnostic laboratories, making the proposed approach practical and accessible.

The project scope includes data preprocessing steps such as handling missing values, encoding categorical variables, feature scaling, and dataset preparation for model training. Multiple supervised machine learning algorithms are implemented and evaluated, including Logistic Regression, Support Vector Machine (SVM), and Random Forest, to analyze their predictive capabilities and comparative performance.

Performance evaluation is carried out using standard classification metrics such as accuracy, precision, recall, and F1-score to ensure a balanced assessment of the models. The focus is not only on prediction accuracy but also on reliability and consistency, which are crucial in medical decision-support systems.

The developed model is intended to function as a supportive diagnostic aid rather than a replacement for professional medical judgment. It is suitable for educational use, research experimentation, and preliminary health screening. The scope of the current work does not include real-time hospital deployment, medical image analysis, or integration with electronic health record systems.

However, the proposed system provides a strong foundation that can be extended in future work by incorporating larger datasets, advanced deep learning techniques, explainable AI methods, and user-friendly interfaces for real-world healthcare applications.

4.0 Proposed Solution:

4.1 Problem Solution

The increasing prevalence of liver-related disorders and the challenges associated with early diagnosis highlight the necessity for intelligent and automated prediction systems. Traditional diagnostic approaches rely heavily on clinical expertise, laboratory analysis, and manual interpretation of multiple biochemical parameters. While effective, these approaches can be time-consuming, resource-intensive, and susceptible to human error, especially when subtle patterns exist across multiple attributes.

To overcome these limitations, this project proposes a machine learning–based liver disease prediction system that leverages historical patient medical data to identify meaningful relationships between clinical features and disease outcomes. The proposed solution adopts a supervised learning approach, where predictive models are trained using labeled patient data indicating the presence or absence of liver disease.

The core objective of the solution is to enable early-stage detection by analyzing routine liver function test parameters that are easily obtainable in most healthcare settings. By processing these parameters simultaneously, the system can detect hidden trends that may not be immediately apparent through conventional analysis.

The proposed model functions as a clinical decision-support tool, assisting healthcare professionals by providing a preliminary assessment. It does not replace medical diagnosis but enhances the diagnostic process by offering fast, consistent, and data-driven predictions..

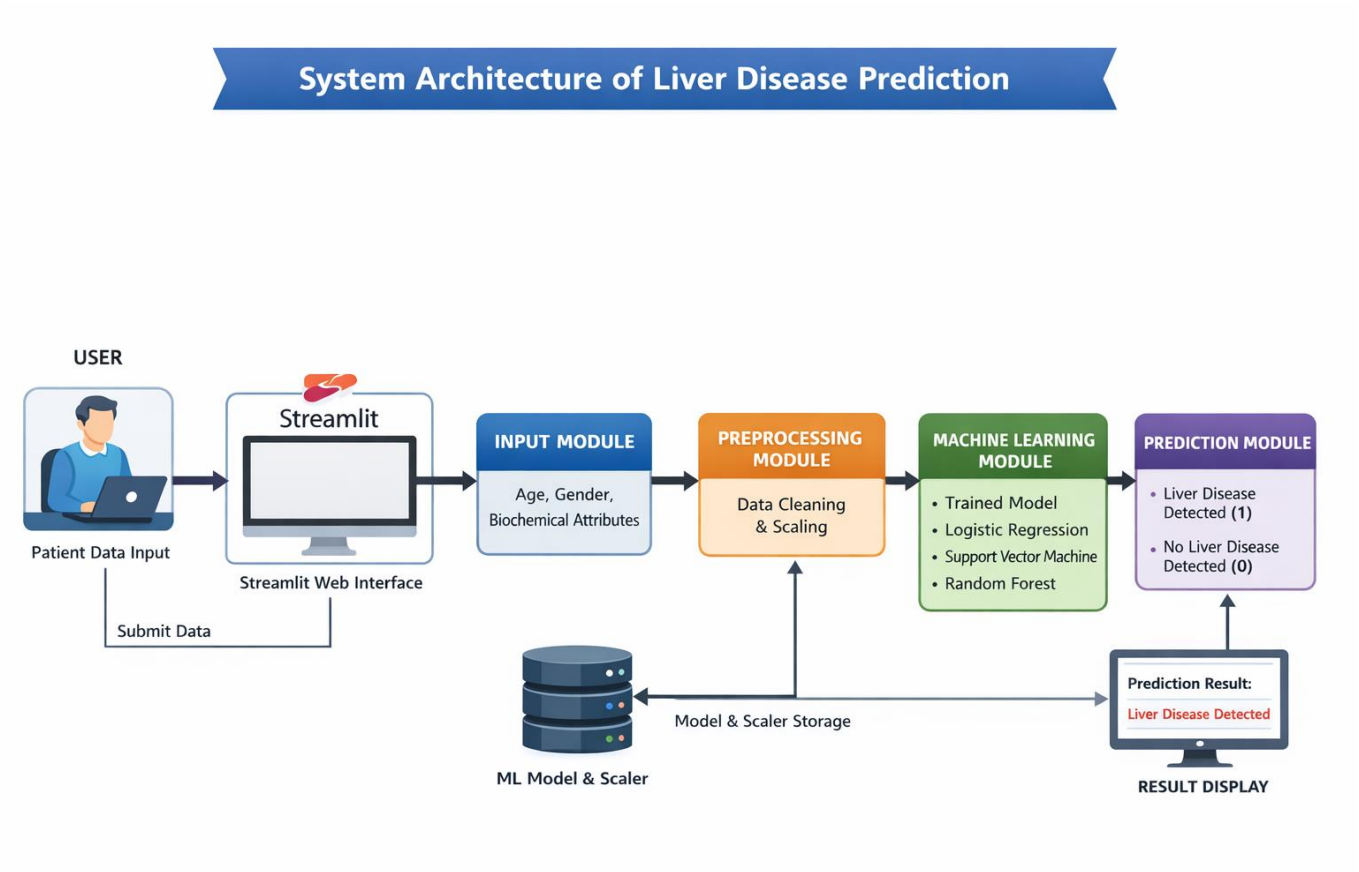

Image 1.1 (Architecture for a system)

4.2 Overview of the Proposed Solution

The proposed solution follows a systematic machine learning pipeline that ensures data reliability, model accuracy, and meaningful predictions. The overall workflow consists of the following stages:

4.2.1. Data Acquisition

The dataset used in this project is the Indian Liver Patient Dataset (ILPD), which contains medical records of patients collected from clinical institutions. The dataset includes demographic and biochemical attributes such as age, gender, bilirubin levels, enzyme measurements, and protein-related values.

4.2.2. Data Preprocessing

Raw medical data often contains inconsistencies, missing values, and varying scales. Therefore, preprocessing is a crucial step in the proposed system. This stage includes:

- Identification and handling of missing values using statistical methods such as median imputation.
- Conversion of categorical variables (e.g., gender) into numerical format.
- Removal of redundant or inconsistent entries.
- Standardization of numerical features to ensure uniform scale across attributes.

4.2.3. Feature Engineering and Selection

Relevant features influencing liver health are selected based on medical significance and statistical importance. This step improves model efficiency, reduces noise, and enhances prediction accuracy.

4.2.4. Model Development

Multiple supervised machine learning classifiers are implemented, including:

- Logistic Regression
- Support Vector Machine (SVM)
- Random Forest Classifier

Each model is trained using the processed dataset to learn patterns distinguishing diseased and non-diseased patients.

4.2.5. Model Evaluation

The trained models are evaluated using standard performance metrics such as accuracy, precision, recall, and F1-score. Comparative analysis is performed to identify the most reliable and consistent classifier.

4.2.6. Prediction System

The best-performing model is deployed as a prediction system capable of accepting new patient data and generating binary output indicating liver disease risk.

	Logistic Regressions	SVM	Random Forest
Accuracy	0.735043	0.709402	0.726496
Precision	0.740741	0.709402	0.757576
Recall	0.963855	1.0	0.903614
F1-score	0.837696	0.83	0.824176

Image 1.2(Evaluation Metrics of Machine Learning Algorithms)

4.3 System Architecture and Working:

The system architecture is designed in a modular manner to ensure clarity, scalability, and ease of implementation. It consists of the following components:

4.3.1 Input Module

This module accepts patient medical parameters such as:

- Age
- Gender
- Total and direct bilirubin
- Alkaline phosphatase
- ALT and AST enzyme levels
- Total proteins
- Albumin and A/G ratio

These values are either extracted from the dataset or provided manually through a user interface.

4.3.2 Preprocessing Module

The preprocessing unit ensures data quality by:

- Handling missing or null values
- Encoding categorical features
- Applying feature scaling using StandardScaler
- Preparing data in the format required by the machine learning model

4.3.3 Machine Learning Module

This module applies trained classification algorithms to analyze the processed data. The model computes decision boundaries based on learned patterns from training samples.

4.3.4 Prediction Module

Based on the model output, the system classifies the patient as:

- Liver Disease Detected (1)
- No Liver Disease Detected (0)

The result is generated instantly and presented in a readable format.

4.3.5 User Interface Module (Optional)

A lightweight interface such as Streamlit can be integrated to allow non-technical users to input patient data and obtain predictions interactively.

4.4 Data Flow Summary

The operational flow of the system can be summarized as follows:

1. Patient medical data is provided as input.
2. Data is validated and cleaned.
3. Missing values are handled and features are scaled.
4. Processed data is passed to the trained machine learning model.
5. The classifier evaluates feature relationships.
6. Prediction outcome is generated.
7. Result is displayed to the user.

This structured flow ensures accuracy, transparency, and consistency in predictions.

4.5 Expected Outcome

The proposed system is expected to deliver the following outcomes:

- Accurate binary classification of liver disease cases
- Improved early detection capability
- Reduced dependency on manual interpretation
- Faster preliminary screening process
- Enhanced understanding of machine learning in healthcare
- Educational and research-level applicability

The system is designed to be scalable and extensible, allowing future integration with advanced algorithms, larger datasets, and real-time clinical systems.

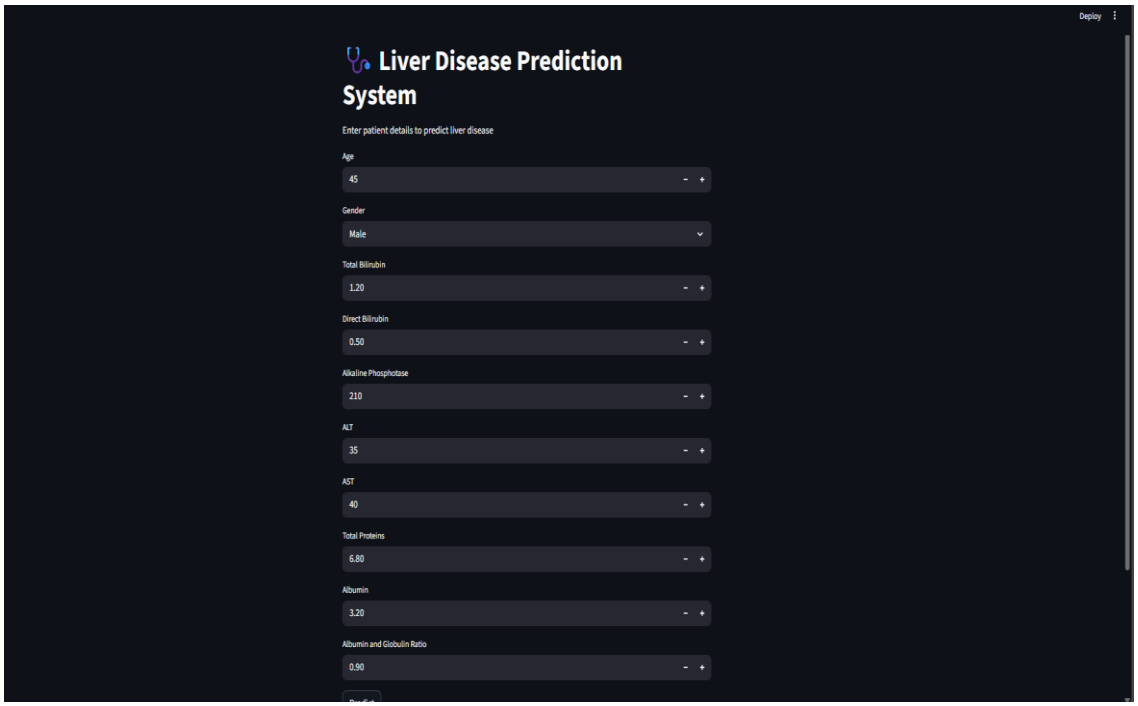

Image 1.3(Streamlit Application for Liver Disease Classification)

5: EXPERIMENTAL SETUP :

5.1 Experimental Setup

The experimental setup defines the environment, tools, dataset, and methodology used to develop and evaluate the liver disease prediction system. This chapter explains how the data was processed, how machine learning models were trained, and how their performance was measured.

All experiments were conducted using the Python programming language due to its extensive support for data analysis and machine learning. Python libraries such as NumPy, Pandas, Scikit-learn, Matplotlib, and Seaborn were used for data preprocessing, model training, and result visualization.

The experiments were performed in a virtual environment using Visual Studio Code with Jupyter Notebook support. This environment allowed step-by-step execution, visualization, and debugging of the machine learning workflow.

The overall experimental process followed these major steps:

1. Dataset loading and exploration
2. Data preprocessing and cleaning
3. Feature encoding and scaling
4. Dataset splitting into training and testing sets
5. Model training using different classifiers
6. Performance evaluation using standard metrics
7. Comparative analysis of results

This structured approach ensured reproducibility, reliability, and fair comparison between different machine learning models.

5.1.1 Dataset Description

The dataset used in this project is the Indian Liver Patient Dataset (ILPD), obtained from the UCI Machine Learning Repository. This dataset is widely used in research studies related to liver disease prediction and serves as a benchmark dataset for evaluating machine learning algorithms in healthcare applications.

Dataset Source

- Repository: UCI Machine Learning Repository
- Dataset Name: Indian Liver Patient Dataset (ILPD)

Dataset Size

- Total instances: 583 patients
- Liver disease patients: 416
- Non-liver disease patients: 167

The dataset exhibits class imbalance, with a higher number of liver disease cases compared to non-disease cases. This characteristic reflects real-world medical scenarios where disease prevalence is often uneven.

Attributes Description:

Attribute	Description
Age	Age of the patient
Gender	Male or Female
Total Bilirubin	Total bilirubin level
Direct Bilirubin	Direct bilirubin level
Alkaline Phosphatase	Enzyme level
Alanine Aminotransferase (ALT)	Liver enzyme
Aspartate Aminotransferase (AST)	Liver enzyme
Total Proteins	Protein concentration
Albumin	Albumin level
Albumin and Globulin Ratio	Protein ratio
Dataset (Target)	1 = Liver disease, 0 = No liver disease

5.1.2 Data Preprocessing:

Before model training, several preprocessing steps were performed:

- **Handling missing values:**
  Missing values in the Albumin and Globulin Ratio feature were filled using median imputation.
- **Categorical encoding:**
  The gender attribute was converted into numerical format (Male → 1, Female → 0).
- **Feature scaling:**
  Numerical features were standardized using the StandardScaler technique to ensure uniform feature distribution.
- **Target variable separation:**
  The dataset column “Dataset” was used as the target variable, while remaining attributes were treated as input features.

5.1.3 Train-Test Split:

To evaluate model performance, the dataset was divided into training and testing subsets:

- **Training set:** 80% of data

- **Testing set:** 20% of data

The split was performed using random sampling to avoid bias and ensure fair model evaluation.

5.1.4 Machine Learning Models Used:

The following supervised learning models were implemented:

1. **Logistic Regression**
2. **Support Vector Machine (SVM)**
3. **Random Forest Classifier**

These models were selected due to their proven effectiveness in medical classification problems and their complementary strengths.

5.1.5 Evaluation Metrics:

Model performance was evaluated using the following metrics:

- **Accuracy:** Overall correctness of prediction
- **Precision:** Reliability of positive predictions
- **Recall:** Ability to identify actual disease cases
- **F1-score:** Balance between precision and recall

These metrics are particularly important in healthcare applications where false negatives can have serious consequences.

5.1.6 Experimental Tools and Technologies

- Programming Language: Python
- IDE: Visual Studio Code
- Environment: Virtual Environment (venv)
- Libraries: Pandas, NumPy, Scikit-learn, Matplotlib

5.2 RESULTS AND PERFORMANCE ANALYSIS

After training and testing the machine learning models, their performance was evaluated using standard classification metrics. The objective of this analysis is to compare different algorithms and identify the most suitable model for liver disease prediction.

The trained models include Logistic Regression, Support Vector Machine (SVM), and Random Forest. Each model was tested on unseen data to measure its generalization ability.

The evaluation metrics used were:

- Accuracy
- Precision
- Recall
- F1-score

These metrics provide a comprehensive understanding of the predictive capability of each model, especially in a medical diagnosis context where minimizing false negatives is crucial.

Comparison of Machine Learning Models

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.735	0.741	0.964	0.838
Support Vector Machine (SVM)	0.709	0.709	1.000	0.830
Random Forest	0.726	0.758	0.904	0.824

Analysis of Results

- **Logistic Regression** achieved the highest F1-score, indicating a strong balance between precision and recall.
- **SVM** demonstrated perfect recall, meaning it successfully identified almost all liver disease cases; however, its precision was lower, resulting in more false positives.
- **Random Forest** performed consistently well with balanced precision and recall but slightly lower overall accuracy compared to Logistic Regression.

From the comparative analysis, Logistic Regression emerged as the most reliable model for this dataset due to its stability, interpretability, and balanced performance.

Confusion Matrix Explanation

A confusion matrix is used to evaluate the performance of a classification model by comparing predicted values with actual values.

It consists of four components:

- **True Positive (TP):**
  Patients correctly predicted as having liver disease.
- **True Negative (TN):**
  Patients correctly predicted as not having liver disease.
- **False Positive (FP):**
  Patients predicted as having liver disease but actually healthy.
- **False Negative (FN):**
  Patients predicted as healthy but actually having liver disease.

In medical applications, **false negatives are the most critical**, as missing a diseased patient can delay treatment. Therefore, models with higher recall are preferred.

The confusion matrix analysis showed that all three models maintained high recall values, making them suitable for preliminary liver disease screening.

6.0 : CONCLUSION & FUTURE SCOPE

6.1 Conclusion

This project successfully demonstrated the application of machine learning techniques for the binary classification of liver disease using clinical and biochemical data. The proposed system utilized supervised learning algorithms to analyze patient medical parameters and predict the presence of liver disease.

Through systematic preprocessing, feature scaling, and model evaluation, multiple classifiers were trained and compared. Among the tested models, Logistic Regression provided the most balanced and reliable performance, achieving a strong F1-score and high recall.

The results confirm that machine learning can play a significant role in early disease detection and decision-support systems in healthcare. The developed model can assist clinicians by providing rapid preliminary screening and reducing dependency on manual interpretation.

Overall, the project validates the effectiveness of data-driven approaches in medical diagnostics and highlights the importance of artificial intelligence in modern healthcare systems.

6.2 Future Scope

Although the proposed system shows promising results, several enhancements can be considered in future work:

- Integration of larger and more diverse medical datasets to improve model generalization.
- Application of advanced deep learning models for improved pattern recognition.
- Implementation of explainable AI techniques to enhance transparency and clinical trust.
- Development of a full-scale web or mobile application for real-time patient screening.
- Integration with hospital information systems and electronic health records.
- Inclusion of imaging data such as ultrasound or CT scans for multi-modal diagnosis.

With these enhancements, the system can evolve into a comprehensive intelligent healthcare solution capable of assisting medical professionals more effectively.

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